O'REILLY LIVE TRAINING

Building AI Apps with Gemini

and Google AI Studio

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AGENDA

What we'll cover

01 **Introduction to Gemini**

The model family and what changed with Gemini 3

02 **Google AI Studio Workflow**

Playground → Build → SDK → Vertex

03 **Five Lessons from Gemini 3**

Multimodality, long context, grounding, generative UI, context engineering

04 **Building & Agentic Workflows**

Demos, the SDK, and where this is going

Section 01

Introduction to Gemini

The model family, the context, and what actually changed with Gemini 3

DEMO

Whiteboard: what is Gemini?

WHAT IS GEMINI

Google's frontier multimodal model family

Native multimodality

One model, one call: text + image + video + audio

Long context

1M+ tokens — used as working memory, not storage

Tool-using by default

Search grounding, code execution, URL context, function calling

Family of sizes

Pro · Flash · Flash-Lite — pick by cost/latency/quality

MODEL PICKER

Which model to use

Stable

Gemini 2.5 Pro

Complex reasoning, production-stable smart tier

Default

Gemini 2.5 Flash

Default for cost/perf — cheap input, fast

Stable

Gemini 2.5 Flash-Lite

High-volume, budget workloads

Preview

Gemini 3.x

Preview era — show what's coming, don't wire production to it

COURSE THESIS

AI work is now input-design work.

Multimodal inputs · curated context · specified output format. Effort moves away from prompt-craft toward designing what the model sees and emits.

Section 02

The AI Studio Workflow

How real builders actually go from idea to deployed prototype

AISTUDIO.GOOGLE.COM

AI Studio at a glance

01	**Playground**	System instructions, model picker, parameters, tool toggles. Where you start.
02	**Build**	Vibe coding: scaffold a React + Node app from a prompt. Annotation Mode FTW.
03	**Stream Realtime**	Live API: voice / webcam / screen-share with low-latency bidirectional audio.
04	**Generate Media**	Imagen, Veo, Lyria RealTime, native speech.
05	**Library**	Saved prompts and projects. ⚠️ no version control — export to Git.

Note: the Tune section is effectively defunct — current Gemini models aren't fine-tunable in the UI.

DEMO

AI Studio walkthrough

Playground · Build · Stream Realtime · Generate Media · Library

THE CANONICAL WORKFLOW

From idea to deployed prototype

1

Ideate in Playground

system instruction · model · tool toggles · iterate

2

Harden the schema

Structured Output toggle + JSON schema = the contract

3

Scaffold in Build

natural-language prompt or fork from the Bundled Apps gallery

4

Get Code → SDK

exported snippet uses google-genai (the new SDK)

5

Deploy or graduate

one-click Cloud Run, or migrate to Vertex AI

PLAYGROUND

Patterns power users actually use

System instruction

Carries role + explicit prohibitions ("do not speculate") — measurably reduces hallucinations

Structured Output toggle

JSON schema, not prose request. Gemini quirk: fields are OPTIONAL by default — tag required ones

Thinking level

Sidebar dial most beginners miss. "high" = materially better SVG/complex reasoning

Tool toggles

Search grounding · Code Execution · URL Context · Function Calling — checkboxes, not separate APIs

Multimodal drop-in

PDFs, videos, screenshots. No preprocessing. Long-audio: compress with ffmpeg if needed

Get Code

Two clicks → working Python/JS/REST snippet using google-genai

POWER-USER MOVES

Logan Kilpatrick's prototyping tricks

Clone-then-build

Screenshot an existing UI · drop it in · type "clone this UI exactly" · iterate from the scaffold rather than from zero.

Single-prompt design exploration

```
"Add a widget so I can click through these styles: [5 options]"
```

Explore variants without spawning N sessions.

Connect Drive early

AI Studio sessions don't persist by default — connect to Google Drive or you'll lose your work.

DEMO

Nano Banana Pro in the Playground

Diagrams, dashboards, infographics, brand collateral

BUILD MODE

"Vibe coding" — what it really is

What you get

- React + Node.js scaffold (Next.js / Angular as options)
- Three-panel IDE: agent reasoning · code · live preview
- Full npm; agent auto-installs deps & self-corrects type errors
- **Annotation Mode**: click on the preview, type the change

Deployment exits

Download ZIP

take to local editor

Push to GitHub

real version control

Deploy to Cloud Run

one-click, secrets stay server-side, scale-to-zero pricing

Reality check: "pair programming with a fast senior dev." Prototyping accelerator, not production deployment.

DEMO

Build mode — fork & deploy

Annotation Mode → Cloud Run

NATIVE TOOLS

What's a UI toggle in AI Studio

- ✓ **Search grounding**
  cite sources, real-time facts
- ✓ **Code Execution**
  sandboxed Python in the loop
- ✓ **URL Context**
  reads URLs in your prompt
- ✓ **Function Calling**
  tools you define
- ✓ **Files API uploads**
  PDFs, videos, audio
- ✓ **Live API**
  voice/video/screen, WebSocket
- ✓ **Imagen / Veo / Lyria**
  image, video, music
- ✓ **MCP**
  Maps example bundled

Almost everything is a checkbox now — convenience vs. programmatic control, not access.

CRITICAL SDK MIGRATION

The SDK trap

DEPRECATED — Nov 30, 2025

```
# old, broken on new features
import google.generativeai as genai

model = genai.GenerativeModel(
    "gemini-1.5-flash"
)
response = model.generate_content(
    "Hi"
)
```

Repo renamed to `deprecated-generative-ai-python`. Won't work for Live API, Veo, Files API v2.

CURRENT — google-genai

```
# new, what AI Studio exports
from google import genai

client = genai.Client()
response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents="Hi",
)
```

Half the tutorials online still show the old pattern. Use `google-genai` only.

DEMO

Get Code → google-genai SDK

From AI Studio prompt to a Python notebook

DATA & PRIVACY

The data-privacy switch

AI Studio (free)

TRAINS ON YOUR DATA

Trains on prompts	Yes
Reviewers see data	Possible
Compliance	—
Setup	Zero

AI Studio (Cloud Billing on)

NO TRAINING

Trains on prompts	No
Reviewers see data	No
Compliance	Limited
Setup	Low

Vertex AI

ZERO DATA RETENTION

Trains on prompts	No (default)
Reviewers see data	No
Compliance	HIPAA / SOC2
Setup	High (GCP/IAM)

Enabling Cloud Billing on the API key flips training off — even within free quotas.

WHEN TO GRADUATE

AI Studio → Vertex AI

Hit 429s on free key

Need an SLA

Org requires IAM auth

Plaintext API keys are out

Region / data residency

Regulated industry

MLOps tooling

Model Registry, Evaluation, Monitoring

Good news: SDK migration is trivial — both sides use unified `google-genai`. One-line endpoint swap.

GOTCHAS

What to watch out for

Dec 2025 quota cliff	Free 2.5 Flash silently cut 250 → 20 RPD (−92%). Design for paid from day one.
No prompt version control	AI Studio is a scratch pad. Export to Git/notebooks immediately.
Gemini 3 instruction regression	Documented in Google's own forum. Many production builders fell back to 2.5 Pro.
Product fragmentation	AI Studio · Vertex AI Studio · Firebase Studio · Gemini CLI · Jules · Opal · Stitch · Colab AI · NotebookLM
"Get Code" is a starting point	Error handling, streaming, auth all need work before production.

Section 03

Five Lessons from Gemini 3

What changed — and how each shift translates to API/AI Studio capability

THE FIVE LESSONS

What actually changed

01

Unified multimodality

one model, one call

02

Long context = working memory

not a storage bin

03

Grounding

turn scattered data into a knowledge base

04

Generative UI

output format = the surface

05

Context > prompt engineering

design what the model sees

+1

Reduced sycophancy

ask for critique, not approval

LESSON 01

Unified multimodality

One model, one call. Video + audio + image processed jointly — not as a pipeline of single-modality tools.

Demo prompt

```
"I just recorded a walkthrough on toggling smart
features in Gmail. Watch the recording and turn it into
a clean step-by-step checklist I can hand to a new
hire."
```

Use cases

- Screen recording → SOP checklist
- User-interview video → "list every pause > 3s and what was on screen"
- Dense PDF → infographic via Nano Banana Pro

DEMO

Multimodality: video → SOP

One model, one call. No frame extraction.

LESSON 02

Long context = working memory

The 1M+ window already existed. What changed is retrieval quality inside it — ~60% better at deep needle-in-haystack reasoning.

STORAGE BIN

"Summarize these documents."

Generic summary. The model glances. Information sits in the window unused.

WORKING MEMORY

"Find the three biggest discrepancies between management strategy in the earnings call and what the financial PDFs actually show."

Cross-source reasoning. The model connects evidence buried in different files.

DEMO

Long context as working memory

Same uploads, two prompts: "summarize" vs "find contradictions"

LESSON 03

Grounding

Workspace, Search, URL Context, your own data — turn scattered digital history into a single queryable knowledge base.

Power combo: URL Context + Search grounding

Search finds relevant URLs · URL Context reads them · lightweight RAG substitute with zero infra. (TDS: "another nail in RAG's coffin" for prototypes.)

Rule of thumb: if you'd normally spend > 10 minutes hunting through emails or docs to reconstruct context, ask Gemini first.

DEMO

Grounding with Search + URL Context

A lightweight RAG substitute, zero infra

LESSON 04

Generative UI

Screen Spot Pro: 11.4% → 72.7%. Output format becomes the artifact — calculators, dashboards, quizzes. The surface designs around the task.

BEFORE

"Compare these 3 platforms."

→ static comparison table

AFTER

"Compare these 3 platforms. Give me a dashboard with a slider for budget and a toggle for region."

→ fully interactive HTML tool with revenue calculator, break-even tab, recommendation quiz

Pro tip: explicitly request the controls you want — sliders, toggles, filters.

DEMO

Generative UI: an interactive HTML tool

Plain text vs. dashboard with sliders and toggles

LESSON 05

Context engineering > prompt engineering

Gemini can guess your tone, format, and length. It can't guess your facts. Spend effort on context, not phrasing.

OLD STYLE

```
"Act as a punchy thought-leadership writer. Voice:
confident but not arrogant. Use 2-3 short paragraphs..."
```

Adjective soup. Generic results.

NEW STYLE

```
"Here are 3 of my VP's previous LinkedIn posts. Based on
these, rewrite this Q4 report as a LinkedIn post."
```

Ground-truth examples. Output sounds like the VP.

DEMO

Context engineering: VP LinkedIn voice

Three previous posts as ground-truth examples

BONUS

Reduced sycophancy

Google explicitly trained Gemini 3 to push back. Use it.

Red-team prompt

```
"Identify storytelling weaknesses and logical contradictions between the sections of this report. Predict the pushback I'd receive from leadership."
```

Ask for critique, not approval. Adopt a critical persona for sharper feedback.

DEMO

Red-team your own deck

Find storytelling weaknesses and contradictions

Section 04

Building & Agentic Workflows

Demos, the SDK, and where this is going

AISTUDIO.GOOGLE.COM/APPS/BUNDLED

Fork from the gallery

Bundled apps are the intended starting point — preview, fork, modify, deploy.

GeoSeeker

Spatial understanding · location reasoning

Maps Travel Planner

Maps + Gemini → trip planning

Structured Data Extractor

Forms / docs → JSON in one click

Nano Banana Pro

Diagrams · dashboards · brand collateral · infographics

DEMO

Bundled apps tour

GeoSeeker · Maps Travel Planner · Structured Data Extractor

NANO BANANA PRO

Visual communication just became machine-native

Diagrams & Layouts

- Flowcharts
- Animated diagrams
- Dashboards
- Editorial spreads
- Blueprints
- Structured one-pagers

Data Visualization

- Visualize academic papers
- Earnings reports → charts
- Representation transforms

Marketing & Brand

- Brand palettes & logos
- Concept art
- Brand collateral

Agentic Workflows

- Agents generating diagrams
- Visual PDF summaries
- Updating onboarding assets

Section 05

Agentic Workflows with Gemini

Whiteboard + code demo: building agents on the same primitives

DEMO

Building agents with Gemini

Whiteboard + code

CLOSING THESIS

Design the inputs.

Specify the outputs. Trust the model.

github.com/EnkrateiaLucca/oreilly-live-trainings · ai.google.dev